

Rohan Tan Bhowmik

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Summary:

Junior at Stanford University pursuing Electrical Engineering and Computer Science with a focus on artificial intelligence. Co-president of Stanford Student Robotics and software lead for the Stanford Autonomous Underwater Vehicle Project, with hands-on experience in applying machine learning to robotics, health, and human-computer interaction. Deeply interested in brain-inspired computational architectures for embodied intelligence.

Education:

Stanford University

B.S., Electrical Engineering and Computer Science, Class of 2027 (GPA: 3.93/4.00)

Publications:

Peer-Reviewed Journal Papers:

1. C. Ding, **R. T. Bhowmik**, "Artificial Intelligence in Multimedia Content Generation: A Review of Audio and Video Synthesis Technique," *Journal of the Society for Information Display*, Invited Paper (accepted), 2025.
2. **R. T. Bhowmik**, Y. S. Jung, J. Aguilera, M. Prunicki, K. Nadeau, "A Multi-Modal Wildfire Prediction and Personalized Early-Warning System Based on a Novel Machine Learning Framework," *Journal of Environmental Management*, **341**, 117908, 2023. ([link](#))
3. **R. T. Bhowmik**, C. Kandathil, S. P. Most, "Automating the Standardized Cosmesis and Health Nasal Outcomes Survey (SCHNOS) Classification with Convolutional Neural Networks," *Facial Plastic Surgery & Aesthetic Medicine*, PMID: 36749153, 2023. ([link](#))
4. **R. T. Bhowmik**, S. P. Most, "A Personalized Respiratory Disease Exacerbation Prediction Technique Based on a Novel Spatio-Temporal Machine Learning Architecture and Local Environmental Sensor Networks," *Electronics*, **11** (16), 2562, 2022. ([link](#))
5. R. Parthasarathy, **R. T. Bhowmik**, "Quantum Optical Convolutional Neural Network: A Novel Image Recognition Framework for Quantum Computing," *IEEE Access*, **9**, 103337, 2021. ([link](#))
6. **R. T. Bhowmik**, Y. S. Jung, J. Aguilera, M. Prunicki, K. Nadeau, "California Wildfire Forecasting Inventory - Extensive Wildfire Prediction Dataset for Artificial Intelligence," *Fire* (under review).

Preprints:

7. **R. T. Bhowmik**, Y. S. Jung, J. Aguilera, M. Prunicki, K. Nadeau, "California Wildfire Forecasting Inventory - Extensive Wildfire Prediction Dataset for Artificial Intelligence," arXiv:2509.11015, 2025.

8. **R. T. Bhowmik**, "A Multi-Modal Wildfire Prediction and Personalized Early-Warning System Based on a Novel Machine Learning Framework," arXiv:2208.09079, 2022.
9. **R. T. Bhowmik**, "A Multi-Modal Respiratory Disease Exacerbation Prediction Technique Based on a Spatio-Temporal Machine Learning Architecture," arXiv:2103.03086, 2021.
10. R. Parthasarathy, **R. T. Bhowmik**, "Quantum Optical Convolutional Neural Network: A Novel Image Recognition Framework for Quantum Computing," arXiv:2012.10812, 2020.

Conferences:

11. **R. T. Bhowmik***, S. Sun*, E. McElhinney, V. Alexopoulos, S. Hickmann, F. Hoffmann, K. Song, R. Sato, L. Skaling, A. Krinos, C. Ogaki, O. Khatib, "3D Reconstruction of Underwater Features in Lake Tahoe with an Autonomous Underwater Vehicle," International Conference of Computer Vision, 2025.
12. **R. T. Bhowmik**, "A Multi-Modal Wildfire Prediction and Personalized Early-Warning System Based on a Novel Machine Learning Framework," Stanford BigEarth Hackathon, 2022.

Experience:

Researcher, IRIS Lab (Oct 2025-Present)

- Working on human-in-the-loop instruction tuning, focusing on human-robot collaborative tasks, under Professor Chelsea Finn.

Co-President, Stanford Student Robotics Club (Sep 2025-Present)

- Promoting student interest and engagement with robotics events and projects on campus, including planning and organizing project showcases, company events, and hackathons.

Software Lead, Stanford Autonomous Underwater Vehicle Project (Oct 2024-Present)

- Led the design and implementation of a ROS 2 based autonomy stack for a new autonomous underwater vehicle, integrating perception, localization, and path-planning.
- Managed a cross-disciplinary software team of ~15 engineers, integrating mechanical, electrical, and research requirements into a unified, field-validated robotics platform for competitions and scientific missions.

Software Engineering Intern, AMD AI Group (Jun 2024-Sep 2025)

- Specialized in machine learning compilers, optimizing model performance across diverse hardware architectures and reducing latency in large-scale deployments.
- Contributed to Wave, a GPU domain-specific language, by designing and extending custom kernels to improve compilation efficiency and boost inference performance on servers.